



PRACTICE Final Exam

Fall 2025

ANSWER KEY

YOUR NAME: _____

YOUR SEMINAR INSTRUCTOR'S NAME: _____

--- This is a double-sided exam ---

Welcome to the Final Exam for Frontiers of Science. Please **write your name and your seminar instructor's name on the cover of this exam**. The exam begins at **2:00 p.m.** and finishes at **5:00 p.m.**, giving you **3 hours** to complete it.

This exam consists of nine questions: four questions covering the Molecules & Life and Climate & Us units, and five Habits and Connections questions. For Molecules & Life and Climate & Us units, there are two questions consisting of three parts each (5 points per part); the Habits and Connections questions consist of just one part (5 points each). All questions add up to 85 points total.

You must answer ALL questions in the exam packet within the empty space that follows each question. Should you need extra space, you may use the empty pages in the back of the exam. If you do so, make sure to clearly indicate below the question where to find your answer.

Feel free to use your calculator as necessary. **To receive full credit, show all your work, and include units where applicable. Be sure to turn in this exam packet at the end of the exam.**

Good luck!

As a guideline for your answers, each question is marked with: [NC] – 'requires no calculation', [SC] – 'requires simple calculations or quantitatively reading a graph' or [C] – 'requires calculations'.

Numerical Constants, Values, and Unit Conversions

R_E (Radius of Earth)	6.4×10^6 m
M_E (Mass of Earth)	6.0×10^{24} kg
M_S (Mass of Sun)	2.0×10^{30} kg
Density of water	1 kg L^{-1}
1 mile	$= 1.61 \text{ km} = 5280 \text{ feet} = 6.34 \times 10^4 \text{ in}$
1 foot	$= 0.305 \text{ m}$
1 metric ton	$= 1000 \text{ kg} = 2,200 \text{ lbs}$
1 lb	$= 0.454 \text{ kg}$
1L	$= 10^{-3} \text{ m}^3 = 0.26 \text{ gallons}$
1 parsec (pc)	$= 3.26 \text{ light years (ly)} = 3.09 \times 10^{16} \text{ m}$
π	$= 3.14$

Metric System Prefixes

T	“tera”	10^{12}
G	“giga”	10^9
M	“mega”	10^6
k	“kilo”	10^3
c	“centi”	10^{-2}
m	“milli”	10^{-3}
μ	“micro”	10^{-6}
n	“nano”	10^{-9}
p	“pico”	10^{-12}

Equations

Distance

$$d = v t$$

d = distance, v = velocity, t = time

Geometry

$$A_{\text{circle}} = \pi r^2; \quad A_{\text{sphere}} = 4\pi r^2$$

$$V_{\text{sphere}} = \frac{4}{3} \pi r^3; \quad V_{\text{cylinder}} = h\pi r^2$$

A = area, r = radius

V = volume, r = radius, h = height

Density

$$D = \frac{M}{V}$$

D = density, M = mass of object, V = volume

Statistics

$$SE \approx \frac{SD}{\sqrt{N}}$$

SE = standard error of the mean, SD = standard deviation,

N = sample size

Probability

For independent events:

$$P(A \text{ AND } B) = P(A) \times P(B)$$

$$P(A \text{ OR } B) = 1 - [1 - P(A)] \times [1 - P(B)] = P(A) + P(B) - P(A) \times P(B)$$

Probability of event A occurring at least once given n opportunities: $1 - [1 - P(A)]^n$

Probability of event A happening given that event B has happened:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{Number of outcomes in which both A and B happen}}{\text{Number of outcomes in which B happens}}$$

MOLECULES AND LIFE

Question 1. [15 points]

1A. [5 points, NC]

As you learned in the lecture, cancer is the second leading cause of death in the United States.

1A. (i) Briefly describe the molecular basis of cancer, highlighting the role of DNA damage and failure of DNA repair.

Cancer occurs when certain cells in the body grow uncontrollably and spread to other body parts. Two types of genes are frequently mutated in cancer – genes that are involved in inhibition of cell growth (tumor suppressors, which make sure that uncontrollable growth is put under control) and in promotion of cell growth (proto-oncogenes). DNA damage to these genes, which can occur from radiation and toxin exposure, for example, can lead to the development of cancer. Mutations to these genes can also be inherited. If DNA repair works properly and eliminates the mutations, then cancer would not occur.

1A. (ii) Consider what happens after DNA in a cell is damaged. Name TWO of the steps of the Central Dogma of Molecular Biology, the associated molecules involved in each of the two steps, and whether or not each step you named is impacted by DNA damage.

The two steps of the Central Dogma are transcription (DNA->RNA) and translation (RNA->protein). The first uses RNA polymerase and the second uses the ribosome. When DNA is damaged, both of these steps are impacted – the transcribed RNA is not the correct/original sequence and the protein product is also not the correct/original protein.

1A. (iii) Which of the following is true about the ribosome? Circle all that apply:

- a. The ribosome facilitates the process of translation.
- b. The ribosome facilitates the process of transcription.
- c. The ribosome is made up of DNA, RNA, and proteins.
- d. The ribosome is made up of RNA and proteins.

Answers a and d are correct.

1B. [5 points, SC]

In a 2023 study, published in the New England Journal of Medicine, scientists reported the results of a phase 3 clinical trial of a cancer drug named seliperatinib to treat medullary thyroid cancer. The authors compared progression-free survival in the seliperatinib group vs. in a group that was treated with one of two other drugs, cabozantinib or vandetanib, based on the physician's choice. Progression-free survival is defined as the time from randomization or initiation of treatment to the occurrence of disease progression or death.

1B. (i) Name two different reasons as to why the authors might have chosen to compare the progression-free survival in the selpercatinib group vs. a group taking one of the other two drugs? What purpose did this second group serve?

The second groups serves as a control group. For a life-threatening disease like cancer, it would be unethical to not provide any treatment to this control group, so they are treated with treatments used previously rather than given a placebo. In addition, if we want to know that the new treatment is both effective and superior to existing treatments, than we need to compare it to them.

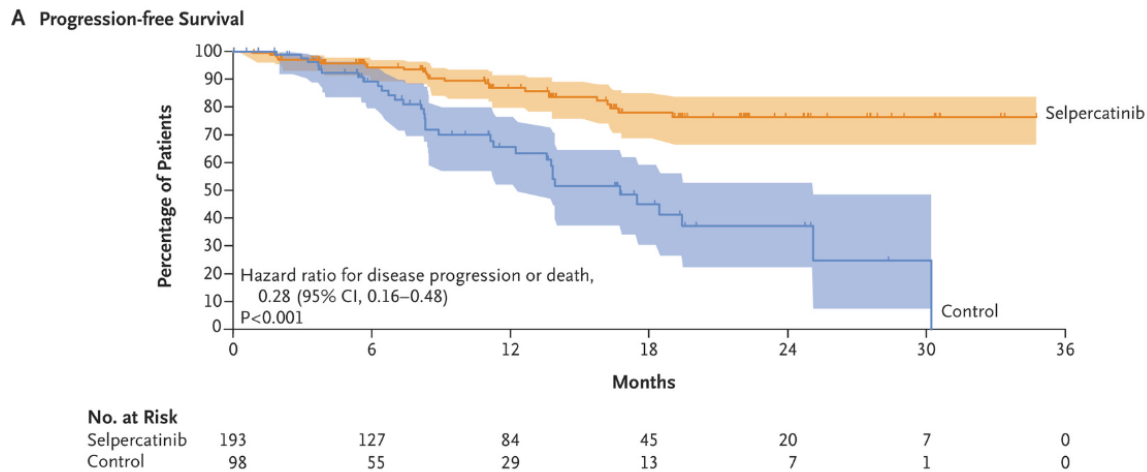


Figure 1. Progression-free survival of Selpercatinib and Control groups.

1B. (ii) Based on the data presented in Figure 1 above, what can you conclude about whether or not selpercatinib is more or less effective than existing treatments? Be sure to explain your answer using your knowledge of statistics.

Selercatinib appears to be more effective than the existing treatments (presented as control here). The hazard ratio for disease progression or death between the experimental and the control groups is statistically significant (lower hazard ratio for the experimental group, which is statistically significant, as $p < 0.001$).

1B. (iii) Based on the data presented in Figure 1, at what point in the trial does the percentage of patients who are progression-free begin to differ between the selpercatinib and the control group? What percentage of patients is progression-free in the selpercatinib group at 33 months after the onset of treatment? Be sure to explain your answer using your knowledge of statistics.

Around 8-9 months after the start of treatment, the progression-free survival of the experimental group begins to differ in a statistically significant way compared to the progression-free survival of the control group. This is because their 95% confidence intervals stop overlapping at that point in the treatment. Nearly three years after the onset of treatment, around 80% of the patients are progression-free.

1C. [5 points, SC]

Figure 2 below shows the data for progression-free survival in subgroups of the population.

B Progression-free Survival in Subgroups

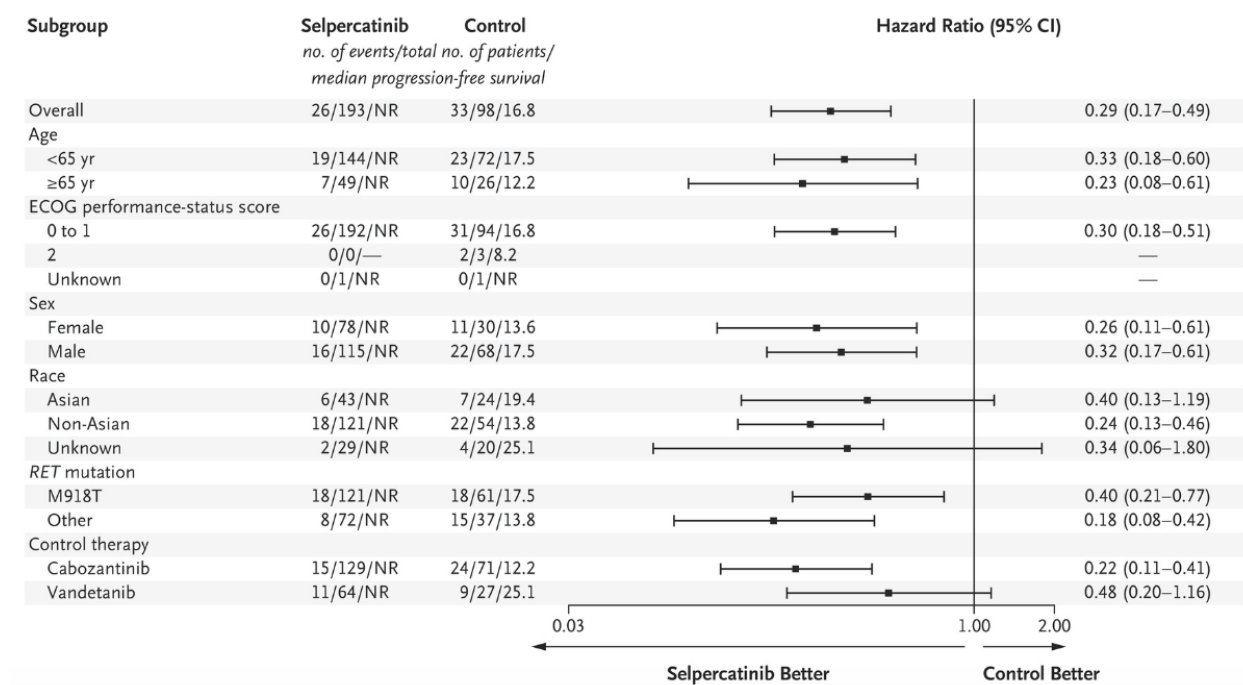


Figure 2. Progression-free Survival in Subgroups

1C. (i) Based on the results shown in Figure 2, is there a statistically significant difference between the comparison of selpercatinib vs. the control for people of different ages (younger vs. equal to or older than 65 years)? Explain using your knowledge of statistics.

No, there is not a statistically significant difference between the two age groups. The 95% confidence intervals for the hazard ratios of these two groups overlap.

1C. (ii) Based on the results shown in Figure 2, which of the two drugs (cabozantinib vs. vandetanib) performs better than the other, if any? Do either or both of the two drugs perform better or worse than selpercatinib and if so, state which one? Explain using your knowledge of statistics.

The 95% confidence intervals overlap for cabozantinib vs. vandetanib, so neither can be said to perform better than the other. However, cabozantinib performs worse than selpercatinib, whereas the 95% confidence interval for vandetanib overlaps with that of a hazard ratio of 1, meaning that it is no worse nor better than selpercatinib.

1C. (iii) Medullary thyroid cancer arises from particular cells of the thyroid gland, called C cells. This cancer is driven by mutations in a proto-oncogene RET kinase and selpercatinib is an RET

kinase inhibitor. **Given what you know about the molecular basis of therapeutics, state what selercatinib is and how it might treat medullary thyroid cancer.**

Selercatinib is a small molecule drug that is capable of binding to the RET kinase in a way that prevents this oncogene from furthering the process of tumor growth and disease progression. Selercatinib likely does this via hydrogen bonding to the RET kinase at just the right location.

Question 2. [15 points]

2A. [5 points, C]

Chronic lymphocytic leukemia (CLL) is a cancer that arises in a type of white blood cells called lymphocytes. A recent study identified 45 common genetic mutations that increase the risk of individuals developing CLL.

2A. (i) Which molecular machine could have made the “mistake” that generated this genetic mutation? Circle all that apply and explain in one sentence.

- a. Ribosome
- b. RNA-dependent RNA polymerase
- c. DNA-dependent RNA polymerase
- d. DNA polymerase

The correct answer is d. DNA polymerase is the only molecular machine mentioned that synthesizes genomic DNA, during replication, and so it is the only one that can generate mutations that are inherited from the original DNA source.

2A. (ii) A genetic engineering lab seeks to determine how well their new CRISPR-Cas9 procedure works when editing lymphocyte cells to remove CLL risk-associated mutations. They aim to target 16 of the 45 identified mutations that are present in a lymphocyte cell. In previous trials, they found that their genome editing procedure could successfully edit a single mutation 90% of the time in a single cell. **Based on this, estimate the probability that they can successfully edit all of the 16 target mutations in 100 lymphocyte cells.**

$$P(\text{one target edited}) = 0.9$$

$$P(\text{all 16 targets edited}) = P(\text{one target edited})^{16} = 0.9^{16} = 0.185 \text{ or } 18.5\%$$

$$P(\text{all 100 cells edited}) = P(\text{all targets in one cell edited})^{100} = 0.185^{100} = 5.21 \times 10^{-74}$$

2A. (iii) Do you think this success rate would make this method a useful therapy for people at risk for CLL? State Yes or No and explain in one sentence.

No; if we assume that all positions in all or most lymphocyte cells must be edited in order to effectively reduce CLL risk in an individual, the answer in part iii suggests that this would be extremely improbable, and therefore this would not be a useful therapy.

Note: the assumption above may not be the case for all diseases, i.e. disease risk may be decreased even when only a subset of risk-associated mutations are edited. Students can be awarded full credit for a “yes” answer if they make this point.

2B. [5 points, NC]

Another disease that involves blood is sickle cell anemia, which is a result of misshapen red blood cells. Although having the disease significantly impairs survival, the sickle cell mutation persists. It is particularly prevalent in areas with high malaria rates as it likely inhibits the malaria parasite's survival within red blood cells. The current hypothesis is that while two copies of the mutated gene can cause serious health issues, individuals with one copy may have some protection against malaria without developing sickle cell disease. If this hypothesis is correct, carriers of the mutation would have a survival advantage in such regions, which could explain why the mutation persists in certain populations.

Williams et al. (2021) investigated whether possessing a single copy of the sickle cell gene provides protection against malaria. The researchers analyzed data from two groups of Kenyan children: one group with one copy of the mutated gene and another group without any mutations.

2B. (i) Does the graph in Figure 3 (next page) support the claim that having one copy of the sickle cell mutation provides significant protection against malaria? Support your answer with relevant statistics.

(i) The figure (next page) indicates that children, aged 2-8, with one copy of the mutation have a statistically significantly lower malaria incidence rate than those with no copies of the mutation, as shown by the 95% CI's of the incidence rate ratio (IRR) consistently not overlapping with 1 (an IRR of 1 indicates no difference in risk). This supports the idea that having one sickle cell gene copy may provide some resistance to malaria. In older children, the difference is not significant due to overlapping 95% CI. Further research is needed to determine whether acquired immunity for malaria increases over time.

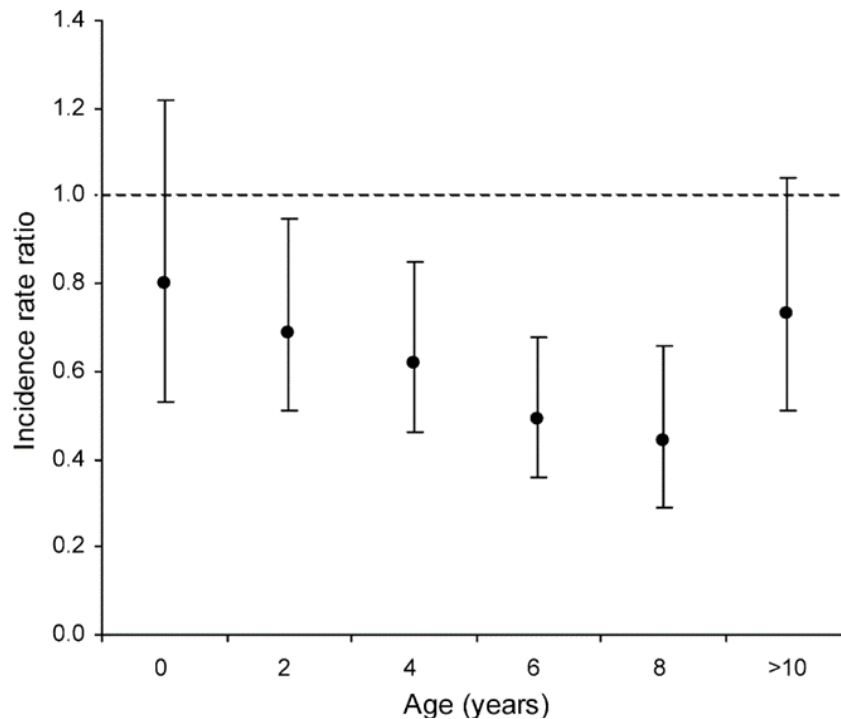


Figure 3. Incidence rate ratio (IRR) of malaria in children with and without the sickle cell mutation. $IRR = \frac{\% \text{ of individuals with one copy of the sickle cell mutation that get malaria}}{\% \text{ of individuals without the sickle cell mutation that get malaria}}$. IRR of 1 indicates an equal risk of developing malaria in both groups. Error bars represent 95% confidence intervals. Adapted from Williams et al. (2005).

2B. (ii) Sickle cell anemia arises due to a single point mutation in the gene for the protein hemoglobin. **Given this, would sickle cell anemia be a good disease candidate for the gene editing technique CRISPR-Cas9?**

Yes, sickle cell anemia is a good disease candidate for gene editing using CRISPR-Cas9. Not only is it a genetic disease, but the fact that it arises due to a single point mutation means that the chance of success is much higher than for diseases involving multiple mutations (or multiple genes).

C. [5 points, C]

2C. (i) Describe how CRISPR-Cas9 was discovered and why it is an attractive tool to use in gene editing.

CRISPR-Cas9 was discovered by researchers trying to understand the immune system of bacteria (it all started with yogurt!). Bacteria use an enzyme called Cas9, which can cut DNA at precise locations, and small pieces of RNA called guide RNAs to attack pathogens, such as viruses. The guide RNA matches a sequence of the viral DNA. Once it binds to the viral DNA, the enzyme Cas9 binds to the guide RNA, and is able to snip out part of the viral DNA. Jennifer Doudna and colleagues used two Cas9 enzymes, making a pair of molecular scissors that could

cut out any segment of DNA. They also figured out a way to add a new sequence into the cut DNA, using the cell's own DNA repair machinery.

To treat sickle cell disease and alleviate its symptoms, it is necessary to increase the number of healthy red blood cells in the patient's bloodstream. One common method for achieving this is through regular blood transfusions, which can temporarily raise the hemoglobin levels in the blood and enhance its ability to carry oxygen.

2C. (ii) What volume of donor blood (in liters) would be required to raise the hemoglobin level of a sickle cell patient from 6 g/dL to 10 g/dL? Assume the following and state any additional assumptions:

- A single bag of blood contains approximately 180 mL of red blood cells, which account for 55% of the volume of blood.
- A hemoglobin increase of 1 g/dL is achieved by transfusing approximately 0.01 bags of blood per kilogram of body weight.

Assumption: the patient weighs 70 kg

Calculate the number of bags of blood required to raise the hemoglobin level from 6 g/dL to 10 g/dL:

Number of bags = (Target hemoglobin level – Initial hemoglobin level) × 0.01 × Body weight.

Number of bags = (10 – 6) × 0.01 bags/kg × 70 kg = 2.80 bags of blood

Calculate the volume of donor blood required:

Volume of donor blood = Number of bags × Volume of packed red blood cells per bag / proportion of red blood cells

Volume of donor blood = 2.8 bags × 180 mL / 0.55 = 466.67 mL = 0.96 liters.

CLIMATE & US

Question 3. [15 points]

3A. [5 points SC]

In lecture, you learned that scientists use proxies to reconstruct past climate or environmental variations.

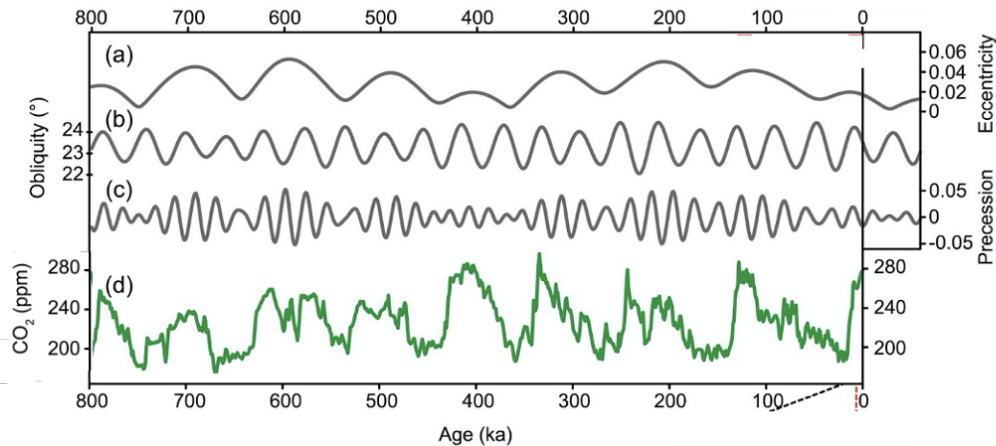


Figure 3. Changes in (a) eccentricity, (b) obliquity, (c) precession, and (d) atmospheric CO₂ over the last 800,000 years. Note that the x-axis, measured in thousands of years (ka), runs forward in time from left to right, and that the y-axis of the CO₂ panel does not extend to current CO₂ concentrations (roughly 415 ppm).

3A. (i) Provide one example of a proxy that is used to reconstruct climate or environmental conditions of the past.

Several answers are possible here. Climate proxies we discussed in lecture/seminar are: tree rings, ice cores, ocean sediment cores, speleothems (cave deposits), corals, and fossils and rocks.

3A. (ii) Describe the specific information that the climate proxy listed in part (i) can provide and whether this proxy can be used to reconstruct climate or environmental conditions over the entire time period shown in Figure 3 above.

Ice cores can provide past CO₂ concentrations (and concentrations of other atmospheric gases), temperature, and precipitation information, and they can span the full time period of Figure 4 (hundreds of thousands of years).

Ocean sediment cores can be used to infer past temperature, precipitation, sea level salinity, etc., and they can also span the full time period of Figure 4 (hundreds of thousands to millions of years).

Tree rings can be used to infer past temperature and precipitation/hydroclimate, but they cannot span the time scale of 800,000 years (max is 12,400 years; most records are on the time scale of hundreds or a couple thousand years).

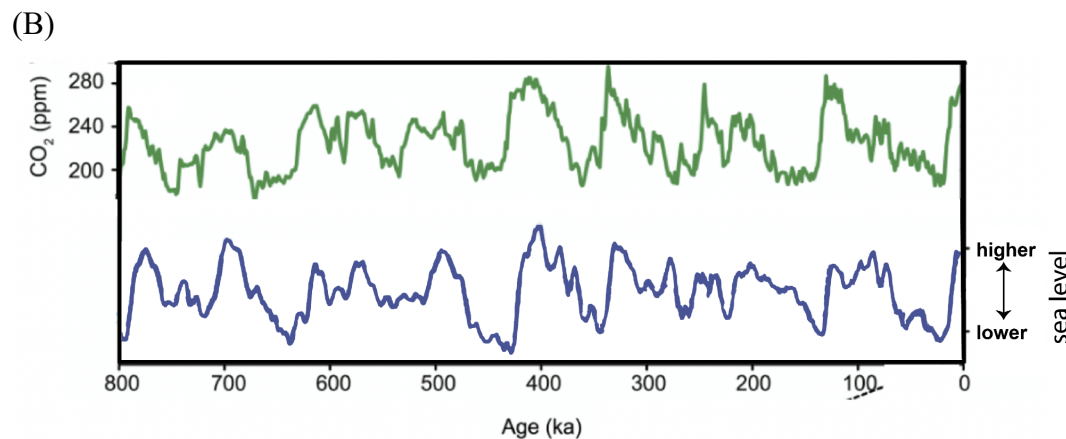
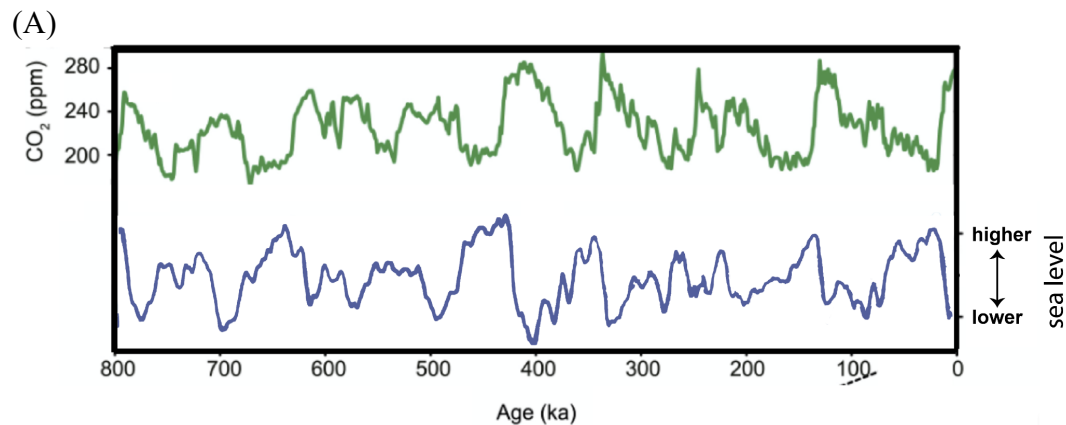
Speleothems provide temperature and precipitation information over thousands of years and do not extend over the full time period of Figure 4 (though few records get close).

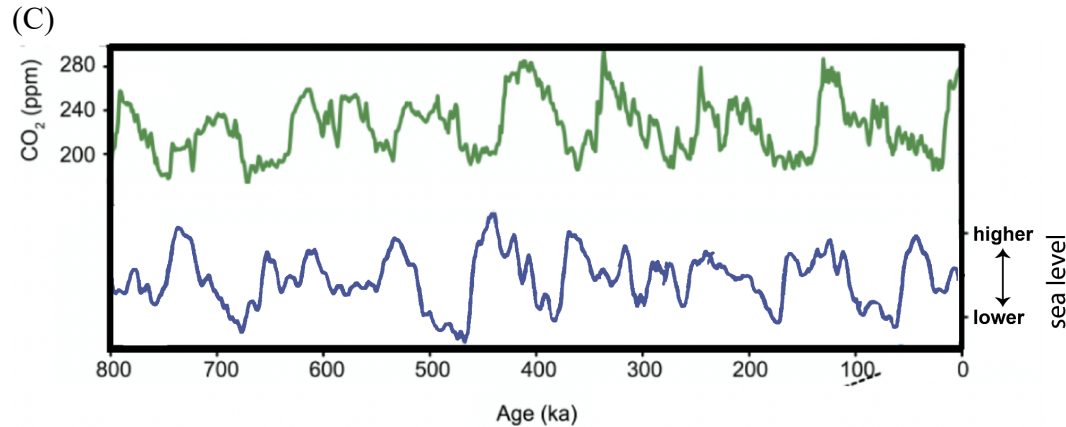
Corals can provide water chemistry and temperature information over decades to centuries and do not extend over the time scale of 800,000 years.

Fossils and rocks can provide past vegetation/climate information (e.g., pollen spores, stomatal density, etc.), and they span the timescale of billions of years.

Note that there are other possible answers that were not discussed in lecture/seminar

3A. (iii) From the three options (A-C; below and next page), select the option that best describes the variation in sea levels from 800,000 years ago to present, based on the reconstructed CO₂ level.





B is correct.

3A. (iv) Are orbital parameters (i.e., eccentricity, precession, and obliquity) relevant to changes in sea level over the past 800,000 years? State yes or no and explain why in 1-2 sentences.

Yes, because CO₂ and temperature fluctuate based on changes in orbital parameters, and sea level tracks these fluctuations.

B. [5 points, C]

Figure 4 below shows some of the major steps of the carbon cycle, where carbon flows into the atmosphere via “carbon sources” and out of the atmosphere via “carbon sinks.”

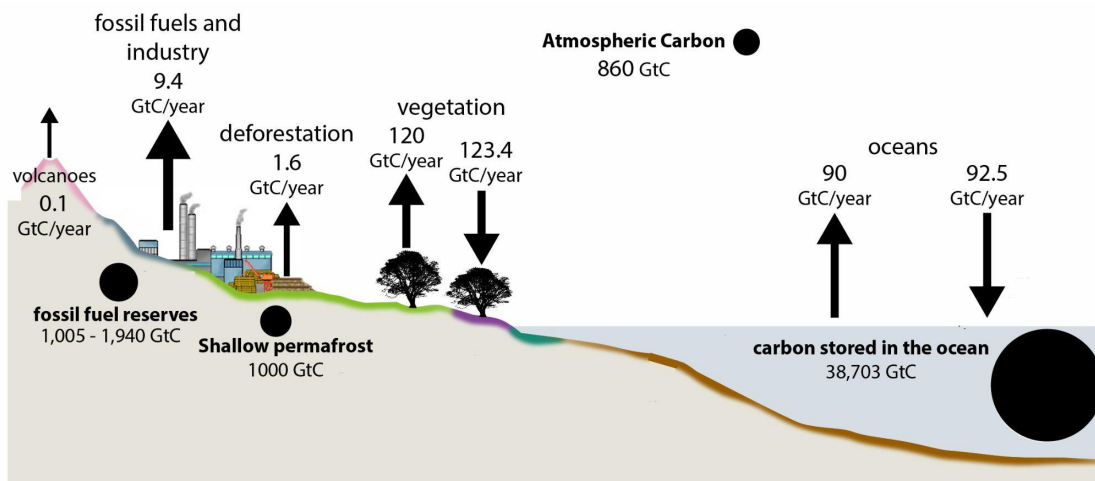


Figure 4. Arrows and accompanying values represent average yearly flux of carbon in metric gigatons of carbon per year, (GtC/year) into and out of the atmosphere over the time period 2010 - 2019. Circles and accompanying values represent stored carbon in gigatons of carbon (GtC) for some major carbon reservoirs on Earth.

3B. (i) Based on Figure 4 (previous page), what was the total net amount of carbon released into the atmosphere during the 2010 - 2019 period (9 years total)? Give your answer in metric tons of carbon and write it using scientific notation.

$$(0.1 \text{ GtC/yea} + 9.4 \text{ GtC/year} + 1.6 \text{ GtC/year} + 120 \text{ GtC/year} + 90 \text{ GtC/year} - 123.4 \text{ GtC/year} - 92.5 \text{ GtC/year}) \times 9 \text{ years} = 5.2 \text{ GtC/year} \times 9 \text{ years} = 46.8 \text{ GtC} = 4.68 \times 10^{10} \text{ tons of C}$$

3B. (ii) A 2019 study by Karjalainen and colleagues estimates that half of the current amount of shallow permafrost will be thawed by the year 2080 even if we moderately reduce greenhouse gas emissions. **Based on this information and Figure 4, what would be the average yearly amount of carbon added to the atmosphere (in GtC/year) from shallow permafrost from 2019 - 2080 (61 years total)?** Assume that the rate of shallow permafrost thaw is constant.

$$500 \text{ GtC} / 61 \text{ years} = 8.2 \text{ GtC/year}$$

3B. (iii) Using Figure 4 and your answers to (i) and (ii) calculate what the total concentration of atmospheric CO₂ (in ppm) would be if half of the shallow permafrost thawed by the year 2080. Use the following:

- (1) All the other aspects of the carbon cycle shown in Figure 5 (sources and sinks) remain the same.
- (2) Assume that all the carbon contained in permafrost is released as CO₂.
- (3) 1 GtCO₂ contains 0.27 GtC.
- (4) The net release of 1 GtCO₂ increases the available concentration of atmospheric CO₂ by 0.127 ppm.
- (5) The concentration of atmospheric CO₂ in 2019 was 412 ppm.

First we have to take into account the new net CO₂ increase per year, taking into account the rate calculated for shallow permafrost thaw in (ii):

$$5.2 \text{ GtC/year} + 8.2 \text{ GtC/year} = 13.4 \text{ GtC/year}$$

Calculate the total increase over 61 years (from 2019 - 2080):
 $61 \text{ years} \times 13.4 \text{ GtC/year} = 817.4 \text{ GtC}$

Use our conversion factor from GtC to GtCO₂

$$817.4 \text{ GtC} \times 1 \text{ GtCO}_2 / 0.27 \text{ GtC} = 3,027.4 \text{ GtCO}_2 \times 0.127 \text{ ppm CO}_2 / 1 \text{ GtCO}_2 = 384.5 \text{ ppm}$$

The concentration in 2019 = 412 ppm, so the concentration of atmospheric CO₂ in 2080 would be 797 ppm if the only change to the current carbon cycle shown in Figure 5 was thawing permafrost!

3B. (iv) Are the assumptions we gave you in part (iii) realistic? Explain why or why not in one sentence.

Answer: It is not reasonable to assume that only permafrost will thaw: the amount of fossil fuel use will change (either increase or decrease depending on legislation, etc.), and the amount of CO₂ able to be absorbed by the ocean will decrease.

3C. [5 points, SC]

Figure 5 below shows the contribution of different anthropogenic factors to the mean global temperature change, based on climate models.

3C. (i) What is the factor that counters climate warming the most and how does this factor cool the earth? Explain in one sentence.

Sulfur dioxide; sulfuric acid particles formed in the atmosphere increase atmospheric reflectivity (albedo), which prevents radiation from reaching the Earth.

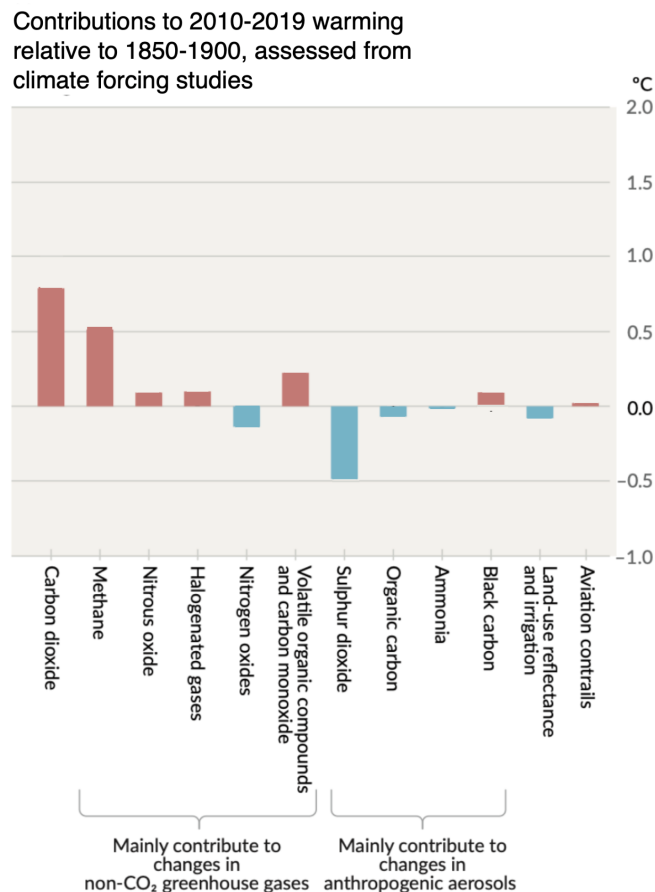


Figure 5. Mean temperature changes, based on the most recent climate models, from individual components of human influence. Estimates account for both direct emissions into the atmosphere and the effect, if any, on other climate forcings.

3C. (ii) When building a realistic global climate model, which of the climate forcings from Figure 5 do scientists need to put into the model? Explain why in one sentence.

All of them; Possible response - Even though some of the forcings like aviation and ammonia are relatively small effects, they are still important in calculating the overall net effect of forcings on climate (and predicting climate warming).

3C. (iii) Consider the temperature change due to CO₂ shown in Figure 5. Has this value been measured directly? State yes or no and explain your answer in one sentence.

No, this value is the result of the modeled impact of atmospheric CO₂ on temperature.

3C. (iv) The Amazon rainforest serves as a carbon sink. However, widespread human-driven deforestation produces carbon dioxide, and increasing deforestation could turn the Amazon into a carbon source. Increased human-driven deforestation would lead to higher levels of carbon dioxide, which would lead to greater amounts of climate warming. As written, is this a feedback loop, and if so, what kind of a feedback loop is it?

Not a feedback loop.

Question 4. [15 points]

4A. [5 points, SC]

Global surface temperatures have increased relative to the time period from 1850-1900 and are predicted to continue to increase.

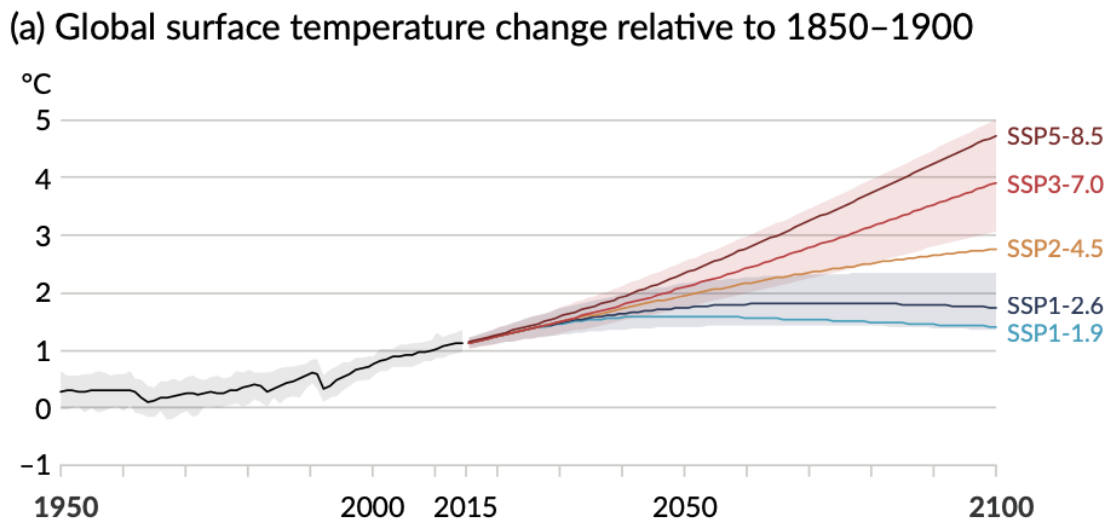


Figure 6. Global surface temperature changes in °C relative to 1850–1900. These changes were obtained by combining global climate model simulations with observationally-based models of past climate. Shades represent very likely ranges for SSP1-2.6 and SSP3-7.0.

4A. (i) Displayed in Figure 6 above are multiple IPCC scenarios. **Why are different scenarios used to make climate change predictions?** Explain in 1-2 sentences.

We have different scenarios for future climate change because 1) they represent different levels of anthropogenic climate forcings (less to more), 2) they represent different socioeconomic pathways, which students could express in a number of different ways about how our social and economic choices globally influence our level of greenhouse gas emission

4A. (ii) The world's largest carbon capture facility was recently built in Iceland. This facility removes 4,000 metric tons of CO₂ from the atmosphere per year and converts it into rock. Fossil fuel emissions currently release 35 Gt CO₂ into the atmosphere per year. **How many facilities with carbon removing capacity equal to the one in Iceland would we need in order to remove the entire annual contribution of fossil fuel emissions from the atmosphere? Would this be realistic?**

$$35 \text{ GtC / year} = (1 \times 10^9 \text{ tons} / 1 \text{ Gton}) \times 35 \text{ Gton} = 3.5 \times 10^{10} \text{ tons of CO}_2 \text{ / year}$$

$$1 \text{ facility} = 4,000 \text{ tons of CO}_2 \text{ / year}$$

$$(3.5 \times 10^{10} \text{ tons of CO}_2 / \text{year}) / (4000 \text{ tons of CO}_2 / \text{year/facility}) = 8.75 \times 10^6 \text{ facilities}$$

Yes, it is unrealistic.

4A. (iii) Other than increasing global carbon sinks, name ONE other way that we could mitigate climate change and explain why this mitigation solution would help.

Lots of possible answers here: using and/or developing alternative energy, reducing global consumption, more carbon-neutral public transportation, voting for leaders that will make progress on climate change legislation and agreements, an array of individual action.

4B. [5 points, SC]

Climate scientists are currently investigating the impacts of global warming. Even at the lower estimates of warming predictions, the latest IPCC report warns: "Climate change will lead to increased frequency, intensity and/or duration of extreme weather events such as heavy rainfall, warm spells and heat events, drought, intense storm surges and associated sea-level rise." The effect of climate change on hurricanes is less clear.

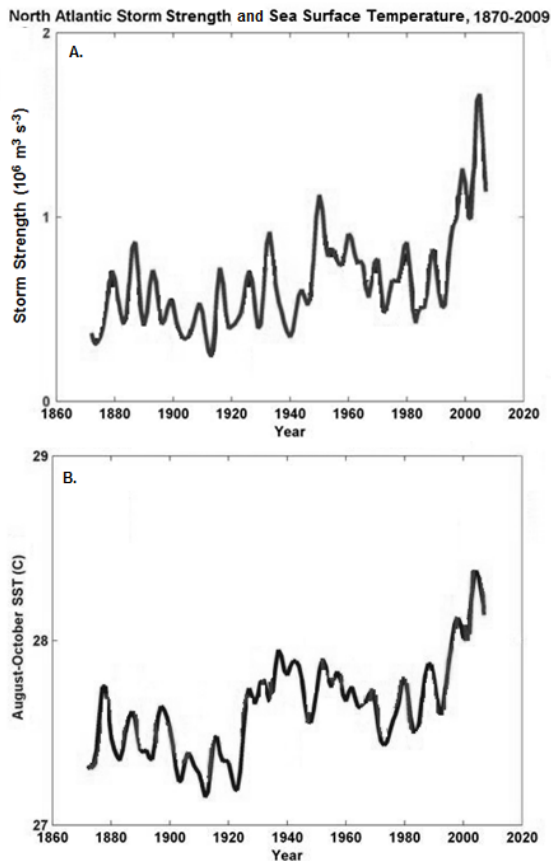


Figure 7A. (Top Panel): Atlantic hurricane strength (calculated from the storm's lifetime maximum wind speed).

Figure modified from Emanuel, K.

<http://eaps4.mit.edu/faculty/Emanuel/publications/>

Figure 7B. (Bottom Panel): Tropical Atlantic sea surface temperature (SST) during hurricane season.

Figure modified from Emanuel, K.

<http://eaps4.mit.edu/faculty/Emanuel/publications/>

Hurricanes obtain their energy from warm surface ocean water. It has been hypothesized that hurricane intensity will increase with global warming.

4B. (i) Do the data shown in Figures 7A and 7B (previous page) support this hypothesis? Explain in 1-2 sentences.

The data could be interpreted as being consistent with the hypothesis because there appears to be a positive correlation between hurricane strength and sea surface temperature.

4B. (ii) Do the data shown in Figures 7A and 7B suggest that anthropogenic global warming is causing changes in the intensity of hurricanes? Explain in 1-2 sentences.

No, they do not because correlation $\neq$ causation.

4B. (iii) In 2007, a paper was published on an increase in the number of observed hurricanes in the North Atlantic over the past century, stating that "improved monitoring in recent years is responsible for most, if not all, of the observed trend in increasing frequency of tropical [hurricanes]". In other words, the reason we're recording more hurricanes is due to our improved ability to observe them using aircraft, radar and satellites. **If the above statement is correct, is this an example of a random or systematic error?** Explain in 1-2 sentences.

Systematic. The earlier data would be systematically biased towards a lower storm count. This would make the change over time seem much larger than it really is.

4C. [5 points, C]

The amount of energy a planet receives depends on its distance from the star and the star's temperature. Based on these two variables alone the average temperature of Earth should be -18°C . **(i) How does Earth reach and maintain a temperature that harbors life?**

Incoming radiation alone is not enough to maintain a constant temperature on a planet, a planet needs albedo to reflect some of the energy and greenhouse gases/atmosphere to hold heat/energy in.

4C. (ii) CarbFix aims to develop technology that turns factory emissions into rock. This process, called carbon sequestration, is possible because CO_2 will mineralize when mixed with water and heat in basalt, a common volcanic rock. The Palisades Sill is a large ($1,750\text{ km}^3$) volcanic flow near NYC that is similar in composition to basalt. **Assuming $\sim 1.3\text{ kg CO}_2$ can be stored in 1 m^3 of basalt, how much CO_2 could the Palisades Sill capture?**

$$\left(1.3 \frac{\text{kg}}{\text{m}^3}\right) \times \left(\frac{1 \times 10^9 \text{ m}^3}{\text{km}^3}\right) = 1.3 \times 10^9 \frac{\text{kg}}{\text{km}^3}$$

$$\left(1.3 \times 10^9 \frac{\text{kg}}{\text{km}^3}\right) \times (1,750 \text{ km}^3) = 2.28 \times 10^{12} \text{ kg CO}_2 \text{ can be stored in the entire Palisades Sill.}$$

4C. (iii) By what factor would storing CO₂ in the Palisades Sill reduce the total atmospheric CO₂? Total atmospheric CO₂ is 5.2x10¹⁸ kg.

$\frac{5.2 \times 10^{18} \text{ kg CO}_2}{(5.2 \times 10^{18} - 2.28 \times 10^{12}) \text{ kg CO}_2} = 1.00000044$ *If the Palisades Sill mineralized all of the possible CO₂ it would reduce the atmospheric CO₂ levels by a factor of 1.00000044.*

The following answers are also acceptable:

$\frac{5.2 \times 10^{18} \text{ kg CO}_2}{2.28 \times 10^{12} \text{ kg CO}_2} = 2.28 \times 10^6$ *If the Palisades Sill mineralized all of the possible CO₂, the amount of CO₂ reduction is a factor of 2.28x10⁶ smaller than the total atmospheric CO₂.*

$\frac{2.28 \times 10^{12} \text{ kg CO}_2}{5.2 \times 10^{18} \text{ kg CO}_2} = 4.4 \times 10^{-7}$ *If the Palisades Sill mineralized all of the possible CO₂, the amount of total atmospheric CO₂ will be reduced by 4.4 x 10⁻⁷%.*

HABITS AND CONNECTIONS

5. [5 points, NC]

Suppose your job is to raise funds for a charitable foundation that supports two of the four scientific fields covered in Frontiers of Science this semester. You talk to potential donors and try to convince them to give money for research. Choose TWO of the units from the course, one from the life sciences (Mind & Brain or Molecules & Life) and one from the physical sciences (Physics & Reality or Climate & Us). For each chosen unit, in one paragraph (100-150 words) describe the best argument that you would make to persuade donors to give to this research area. Each of your two arguments should use one specific example of past accomplishments and one future prospect discussed in the lectures, readings, and/or seminars.

Many possible answers.

6. [5 points, SC]

Quarks are fundamental elementary particles, which combine to form protons and neutrons. Figure 8 below shows measurements of the mass of the top quark over time. **Describe how the measurements of the mass of the top quark have changed over time. Explain in 1-2 sentences, using the concepts of random and systematic error.**

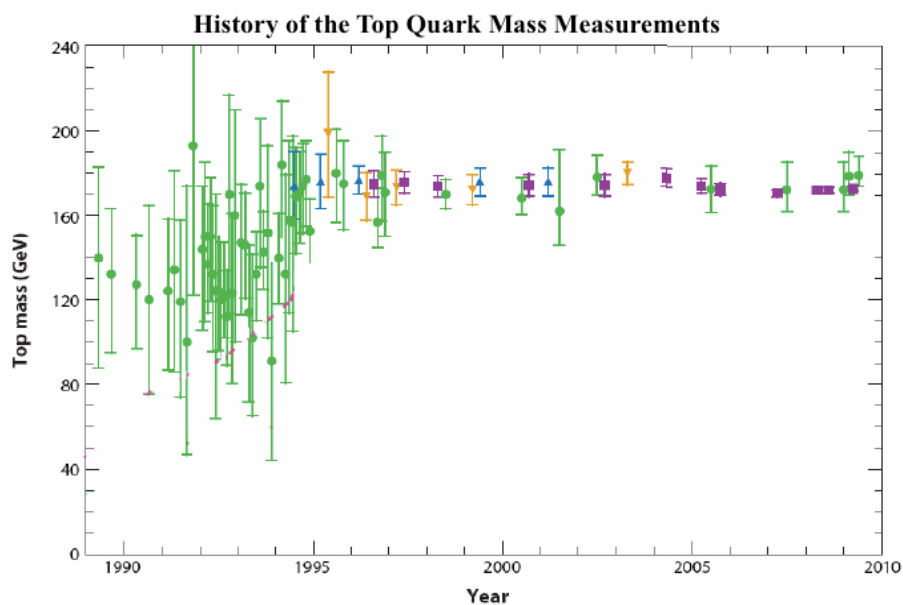


Figure 8. Measurements of the top quark mass by different experiments over time. Error bars represent ± 1 Standard Deviation (SD). Assume that N is approximately equal for the different experiments, and is greater than 10,000. Figure taken from Galtieri et al. (2012).

Measurements appear to have suffered from both large random errors (big error bars) and large systematic errors (mean values at ~ 140 GeV vs. ~ 180 GeV later on) in the beginning experiments (1990-1995), settling down completely by around year 2000. Random errors initially affected precision, and systematic errors affected accuracy – both types of errors appear to have been largely eliminated over time.

7. [5 points, C]

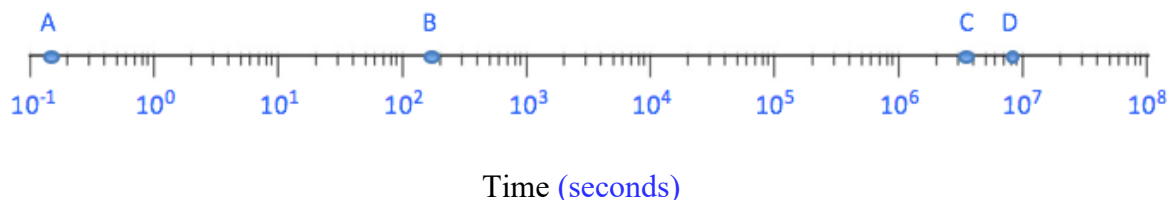
Plot the times listed below (A-D) on the log-scale provided below. Be sure to label the axis and the major ticks on the axis first, then plot all the points with their corresponding letter designations. Show your calculations on the next page

- A. Average reaction time in humans to touch stimulus (150 ms)
- B. Time after the Big Bang when the nuclei of simple elements, such as hydrogen and helium, were formed (3 minutes)
- C. Lifespan of Aedes Aegypti, the mosquito responsible for the spread of the Zika virus (6 weeks)
- D. Interruption of photosynthesis due to the K-T asteroid impact (3 months)



Convert everything to the same unit, e.g. seconds:

- A. Average reaction time in humans to touch stimulus (150 ms) = $1.5 \times 10^{-1} s$
- B. Time after the Big Bang when the nuclei of simple elements, such as hydrogen and helium, were formed (3 minutes) = $180 s$
- C. Lifespan of Aedes Aegypti, the mosquito responsible for the spread of the Zika virus (6 weeks) = $6 \times 7 \times 24 \times 60 \times 60 = 3.6 \times 10^6 s$
- D. Interruption of photosynthesis due to the K-T asteroid impact (3 months) = $1/4 \times 365 \times 24 \times 60 \times 60 = 8 \times 10^6 s$



8. [5 points, C]

Estimate the number of human cells in the human body. State all your assumptions and show all your work. Assume that a cell can be approximated as a cube with a side equal to $20\text{ }\mu\text{m}$.

We can begin by roughly estimating the volume of the human body as fitting in a $1.8\text{ m} \times 0.5\text{ m} \times 1\text{ m}$ box, giving a volume of 0.9 m^3 . We can then say that the human body would fill about $2/3$ of the box, or 0.6 m^3 .

An average human cell is approximately 20 microns in dimension so this tells us that a cell takes up a volume of approximately $(2 \times 10^{-5}\text{ m})^3 = 8 \times 10^{-15}\text{ m}^3$.

Dividing the volume of the body by the volume of a single cell will give us the total number of cells in the body:
 $N = (0.6\text{ m}^3) / (8 \times 10^{-15}\text{ m}^3) = 7.5 \times 10^{13}$ or roughly 75 trillion cells.

Recent estimates put the actual number at 37.3 trillion.
<http://www.tandfonline.com/doi/abs/10.3109/03014460.2013.807878>

9. [5 points, C]

The city of Frontierstown holds a random draw lottery (with odds of winning similar to the current New York State lottery). The chances of a single ticket winning different prizes are listed in the table below. Winning each prize is an independent event (each ticket could, theoretically, win all four prizes).

Prize	\$1,000,000	\$50,000	\$100	\$7
Probability of Winning	1 in 12,000,000	1 in 900,000	1 in 10,000	1 in 300

(i) If you buy one ticket, what is the probability that you will win at least one prize?

$P(\text{at least one prize}) = 1 - (1 - 1/12,000,000)(1 - 1/900,000)(1 - 1/10,000)(1 - 1/300) = 3.43 \times 10^{-3}$ or 1 in 291

(ii) If you buy 50 tickets, what is the probability that you will win at least one prize?

$P(\text{at least one win out of 50}) = 1 - (1 - 3.43 \times 10^{-3})^{50} = 16\%$